WORKED SOLUTION

STEP I 2012, Question 3

Pure • Inequalities from Areas

STEP I 2012, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s1-q3

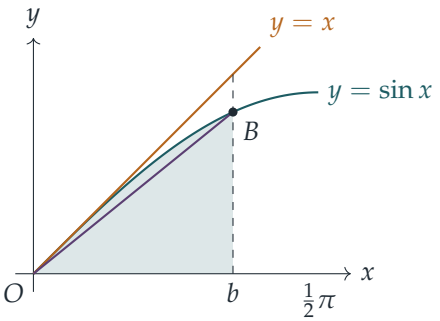
The key idea

A curve that bends one way all the time lies on one side of each of its tangents and on the other side of each of its chords. So the area under the curve is trapped between the area under a tangent and the area under a chord, and both of those are just triangles or trapezia. Write each area down exactly and the inequalities fall out; the only real work is deciding which tangent and which chord to use.

Solution

Part (i)

On $0 < x < \frac{1}{2}\pi$ we have $\dfrac{\mathrm{d}^2}{\mathrm{d}x^2}\sin x = -\sin x < 0$, so the curve $y = \sin x$ is concave there: it bends downwards. The tangent at the origin has gradient $\cos 0 = 1$, so it is the line $y = x$. The chord from O to $B = (b, \sin b)$ is the line $y = \dfrac{\sin b}{b}\,x$.



Concavity means that between O and B the curve lies strictly *above* the chord, and for

$x > 0$ it lies strictly *below* the tangent at O. So, for $0 < x < b$,

$$\frac{\sin b}{b}\, x \;<\; \sin x \;<\; x.$$

Now compare the three areas over $0 \le x \le b$.

- Under the curve (shaded): $\displaystyle\int_0^b \sin x \,\mathrm{d}x = 1 - \cos b$.
- Under the tangent: a right-angled triangle with base b and height b, area $\frac{1}{2}b^2$.
- Under the chord: a triangle with base b and height $\sin b$, area $\frac{1}{2}b\sin b$.

Because the curve is strictly between the two lines on $0 < x < b$ (and all three are continuous), the areas satisfy strict inequalities:

$$\tfrac{1}{2}b\sin b \;<\; 1-\cos b \;<\; \tfrac{1}{2}b^2.$$

The right-hand inequality rearranges to $\cos b > 1 - \frac{1}{2}b^2$ and the left-hand one to $\cos b < 1 - \frac{1}{2}b\sin b$. Together,

$$1 - \tfrac{1}{2}b^2 < \cos b < 1 - \tfrac{1}{2}b\sin b,$$

as required.

Part (ii)

Write $k = \ln a$; since $a > 1$, $k > 0$, and $a^x = \mathrm{e}^{kx}$. Then

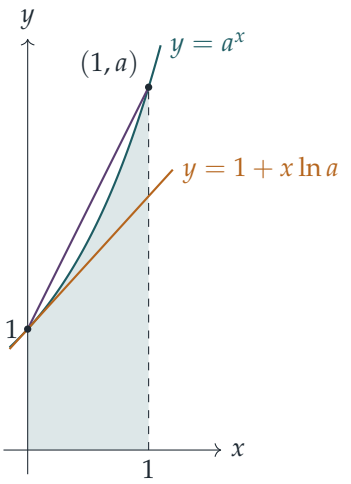
$$\frac{\mathrm{d}}{\mathrm{d}x}a^x = k\,\mathrm{e}^{kx}, \qquad \frac{\mathrm{d}^2}{\mathrm{d}x^2}a^x = k^2\mathrm{e}^{kx} > 0,$$

so this curve is convex: it bends upwards. Now the roles swap: the curve lies strictly *below* its chords and strictly *above* its tangents (away from the point of contact).

? **Ask yourself** Over which interval should we compare areas?

The answer is to involve $\ln a$ and a, but nothing else. The interval $0 \le x \le 1$ does exactly that: the endpoints of the curve are $(0,1)$ and $(1,a)$, and the area under the curve is

$$\int_0^1 \mathrm{e}^{kx}\,\mathrm{d}x = \frac{\mathrm{e}^k - 1}{k} = \frac{a-1}{\ln a}.$$



Lower bound (chord). The chord from $(0,1)$ to $(1,a)$ lies above the curve on $0<x<1$, and the region under it is a trapezium of width 1 with parallel sides 1 and a, of area $\frac{1}{2}(1+a)$. Hence

$$\frac{a-1}{\ln a} < \frac{a+1}{2} \quad\Longrightarrow\quad \ln a > \frac{2(a-1)}{a+1},$$

where we multiplied through by the positive quantity $2\ln a/(a+1)$, so the inequality keeps its direction.

Upper bound (tangent). The tangent at $(0,1)$ has gradient $k=\ln a$, so it is $y=1+kx$. It lies below the curve for $x>0$, and the area under it on $[0,1]$ is a trapezium with parallel sides 1 and $1+k$, of area $1+\frac{1}{2}k$. Hence

$$1+\tfrac{1}{2}k < \frac{a-1}{k}.$$

Multiplying by $2k>0$:

$$2k+k^2 < 2a-2 \quad\Longleftrightarrow\quad k^2+2k+1 < 2a-1 \quad\Longleftrightarrow\quad (k+1)^2 < 2a-1.$$

Ask yourself Can we just take square roots?

Yes, but only because both sides are positive: $2a-1>1>0$, and $k+1>0$ since $k>0$. For positive numbers, $u^2<v^2$ gives $u<v$, so $k+1<\sqrt{2a-1}$, that is,

$$\frac{2(a-1)}{a+1} < \ln a < -1+\sqrt{2a-1},$$

which completes the proof.

Checks

- Part (i) with $b=1$: $1-\frac{1}{2}=0.5$, $\cos 1\approx 0.5403$, and $1-\frac{1}{2}\sin 1\approx 0.5793$. In order, as claimed.

- Part (ii) with $a = 2$: $\frac{2}{3} \approx 0.6667$, $\ln 2 \approx 0.6931$, and $\sqrt{3} - 1 \approx 0.7321$. Again in order.
- As $a \to 1$ all three expressions in part (ii) tend to 0, and they squeeze together: at $a = 1.01$ they agree to five decimal places. That makes sense, because over a short range a smooth curve is almost straight.
- We checked both chains of inequalities at thousands of random values, and the area formulas by Simpson's rule.

Common slips

- Getting the sides the wrong way round: the sine curve is concave (below its tangent, above its chord), while a^x is convex (the opposite). Justify this with the second derivative rather than the sketch.
- Forgetting that $\ln a > 0$ when multiplying or dividing an inequality by it.
- Square-rooting $(k+1)^2 < 2a - 1$ without checking that $k + 1 > 0$.

Beyond STEP

The chord bounds are a case of a general principle: for a convex function, the average value over an interval lies between the value at the midpoint and the average of the endpoint values (the Hermite–Hadamard inequality; using the tangent at the midpoint instead of at an end gives an even sharper lower bound). The lower bound in part (ii) says that the logarithmic mean $(a-1)/\ln a$ of a and 1 is less than their arithmetic mean, a fact that turns up throughout analysis and in physics. Bartle and Sherbert's *Introduction to Real Analysis* treats convex functions and integral estimates carefully.

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